BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Chemistry (Objective) Group - I

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** 5483

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Normal human body temperature is in °C:

(A) 0°
(B) 37°
(C) 98°
(D) 100°

سیلسیس سکیل پر نارمل انسانی جسم کا درجہ حرارت ہے:

Q2. A homogeneous mixture of two or more substances is called:

(A) Solution
(B) Solvent
(C) Solute
(D) Aqueous

دو یا دو سے زیادہ اشیاء کا ہوموجینیس مکسچر کہلاتا ہے:

(A) سلوشن
(B) سالونٹ
(C) سولوٹ
(D) ایکوئس

Q3. A solution which is formed by dissolving a substance in water is called:

(A) Saturated
(B) Solvent
(C) Solute
(D) Aqueous

ایسا سلوشن جو کسی شے کو پانی میں حل کرنے سے وجود میں آئے کہلاتا ہے:

(A) سیچوریٹڈ
(B) سالونٹ
(C) سولوٹ
(D) ایکوئس

Q4. The addition of oxygen or removal of hydrogen during a chemical reaction is defined as:

(A) Oxidation
(B) Reduction
(C) Electricity
(D) Electrolyte

کسی کیمیکل ری ایکشن کے دوران آکسیجن کے اخراج کے عمل کو کہتے ہیں:

(A) آکسیڈیشن
(B) ریڈکشن
(C) الیکٹریسٹی
(D) الیکٹرولائٹ

Q5. Slow and continuous eating away of a metal by the surrounding medium is called:

(A) Paint
(B) Alloy
(C) Corrosion
(D) Stains

کسی میٹل کا ارد گرد کے میڈیم کی وجہ سے کروڑوں جانے کا نام کہلاتا ہے:

(A) پینٹ
(B) [illegible in source scan]
(C) کروژن
(D) [illegible in source scan]

Q6. It is a very reactive metal:

(A) Zinc
(B) Tin
(C) Lead
(D) Sodium

یہ بہت ری ایکٹیو میٹل ہے:

(A) زنک
(B) ٹن
(C) لیڈ
(D) سوڈیم

Q7. A piece of matter in pure form is termed as:

(A) Atom
(B) Substance
(C) Mixture
(D) Element

مادے کا ٹکڑا جو اپنی خالص حالت میں پایا جاتا ہے:

(A) ایٹم
[illegible in source scan] **(B)**
(C) مکسچر
(D) ایلیمنٹ

Q8. N-Shell can accommodate electrons:

(A) 8
(B) 18
(C) 32
(D) 50

N شیل میں الیکٹرونز سما سکتے ہیں:

Q9. In periodic table horizontal rows are called:

(A) Groups
(B) Periods
(C) Metals
(D) Non-metals

پیریاڈک ٹیبل میں افقی قطاریں کہلاتی ہیں:

(A) گروپس
(B) پیریڈز
(C) میٹلز
(D) نان میٹلز

Q10. The size of iodine atom is bigger than of chlorine. So, its electron affinity is:

(A) Greater
(B) Less
(C) Equal
(D) Negative

آئیوڈین ایٹم کا سائز کلورین کے ایٹم سے بڑا ہے۔ پس اس کی الیکٹرون افینٹی ہے:

(A) زیادہ
(B) کم
(C) برابر
(D) نیگیٹو

Q11. The valence electrons which are involved in chemical bonding are termed as:

(A) Attractive forces
(B) Repulsive forces
(C) Bonding electrons
(D) System energy

کیمیکل بانڈنگ میں حصہ لینے والے ویلنس الیکٹرونز کو کہا جاتا ہے:

(A) اٹریکٹیو فورسز
(B) ریپلسو فورسز
(C) بانڈنگ الیکٹرونز
(D) سسٹم انرجی

Q12. The non-bonded electron pair available on an atom is called:

(A) Ion
(B) Single bond
(C) Lone pair
(D) Radical

نان بانڈڈ الیکٹرون جوڑا جو ایک ایٹم پر موجود ہوتا ہے کہلاتا ہے:

(A) آئن
(B) سنگل بانڈ
(C) لون پیئر
(D) ریڈیکل

Chemistry (Subjective) Group - I

Time Allowed: 01:45 Hours | **Maximum Marks:** 48 | **Paper Code:** 5483

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) Define homoatomic molecule and give example.

(ii) What is relationship between Avogadro's number and mole?

(iii) Define empirical formula with example.

(iv) Write two observations made by Rutherford from his experiment.

(v) Write the electronic configuration of Al^{3+}. How many electrons are present in its outermost shell?

(vi) Why electron affinity increases in a period?

(vii) What are the trends of ionization energy in the periodic table?

(viii) How many elements are in the first period of periodic table? Write their names also.

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) Differentiate between octet and duplet rule.

(ii) Define hydrogen bonding.

(iii) Why does group-1 elements prefer to combine with group-17?

(iv) What is the difference between evaporation and condensation?

(v) What is absolute temperature scale and absolute zero?

(vi) Define solution with an example.

(vii) What do you mean by percentage mass/mass (% m/m)?

(viii) How is 1M solution of NaOH prepared?

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) Differentiate between oxidation and reduction.

(ii) What do you mean by strong electrolytes? Give two examples.

(iii) Define rusting. Write its important condition.

(iv) Write the chemical reaction of electroplating of silver.

(v) What do you mean by metals?

(vi) Write any two important chemical properties of metals.

(vii) Write any two uses of silver.

(viii) What are halogens? Give any two examples.

Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 09 نمبر ہیں۔

Q5(a). (Marks 05) Write the postulates of Bohr's Atomic Theory.

Q5(b). (Marks 04) Define molecule. Write its types with examples.

Q6(a). (Marks 05) Define ionic bond and explain it with an example.

Q6(b). (Marks 04) State Boyle's Law and give its mathematical explanation.

Q7(a). (Marks 05) Define corrosion. How corrosion is prevented through zinc coating?

Q7(b). (Marks 04) Define solubility. Also write any three factors which affect solubility.

GROUP – II

Chemistry (Objective) Group - II

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** 5482

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Which compound is an example of homoatomic molecule?

(A) CO_2 (B) H_2SO_4

(C) CH_4 (D) O_3

کون سا کمپاؤنڈ ہومو ایٹامک مالیکیول کی مثال ہے؟

Q2. The P-Sub-shell can accommodate the number of electrons:

(A) 4 (B) 6

(C) 3 (D) 8

P-سب شیل میں زیادہ سے زیادہ کتنے الیکٹرون سما سکتے ہیں؟

Q3. The valency of the elements of group-2 is:

(A) 4 (B) 1

(C) 3 (D) 2

دوسرے گروپ کے عناصر کی ویلنسی ہوتی ہے:

Q4. Halogens belong to group: _____

(A) 15 (B) 16

(C) 18 (D) 17

ہیلوجنز کا تعلق کس گروپ سے ہے؟

Q5. Which is a non-polar molecule?

(A) HCl (B) NH_3

(C) H_2O (D) H_2

کون سا نان پولر مالیکیول ہے؟

Q6. Hydrogen bonding is:

(A) Intermolecular force
(B) Ionic force
(C) Covalent force
(D) Metallic force

ہائیڈروجن بانڈنگ کہلاتی ہے:

(A) انٹرمالیکیولر فورس
(B) آئیونک فورس
(C) کوویلنٹ فورس
(D) میٹالک فورس

Q7. The equation of Charles' Law is:

(A) P = VT
(B) PV = k
(C) V/T = k
(D) T/P

چارلس کے قانون کی مساوات ہے:

Q8. Which is soluble in water?

(A) Naphthalene
(B) Paints
(C) Grease
(D) Sugar

کون سا پانی میں حل ہو جاتا ہے؟

(A) نیفتھیلین
(B) پینٹس
(C) گریس
(D) شوگر

Q9. Molarity is the number of moles of solute dissolved in:

(A) 1kg of solution
(B) 100g of solvent
(C) $1dm^3$ of solvent
(D) $1dm^3$ of solution

مولیرٹی کسی سلوشن کے مولز کی تعداد ہے جو حل شدہ ہو:

(A) سلوشن کے 1 کلوگرام میں
(B) سالونٹ کے 100 گرام میں
(C) سالونٹ کے $1dm^3$ میں
(D) سلوشن کے $1dm^3$ میں

Q10. The oxidation number of chlorin in $KClO_3$ is:

(A) +1
(B) +4
(C) +5
(D) +7

$KClO_3$ میں کلورین کا آکسیڈیشن نمبر ہوتا ہے:

Q11. The example of non-electrolyte is:

(A) NaOH
(B) HCl
(C) H_2SO_4
(D) C_6H_6

کون سا نان الیکٹرولائٹ ہے؟

Q12. Which will not react with ailute HCl ?

(A) Sodium
(B) Potassium
(C) Calcium
(D) Carbon

کون سا HCl کے ساتھ تھوڑی سی ری ایکٹ نہیں کرتا؟

(A) سوڈیم
(B) پوٹاشیم
(C) کیلشیم
(D) کاربن

Chemistry (Subjective) Group - II

Time Allowed: 01:45 Hours | **Maximum Marks:** 48 | **Paper Code:** 5482

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) Define element. How element is represented?

(ii) An atom having A = 238 and Z = 92. Determine number of protons and neutrons.

(iii) Define triatomic molecule with one example.

(iv) What is electronic configuration?

(v) Write two observations of Rutherford's experiment.

(vi) Define ionization energy and write its trends in period.

(vii) Write the trend of atomic radius in periods.

(viii) What is the first ionization energy?

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) Define metallic bond.

(ii) Differentiate between donor atom and acceptor atom.

(iii) Write four types of chemical bond.

(iv) Define vapour pressure.

(v) State Charles' Law.

(vi) Define unsaturated solution.

(vii) What do you mean by percentage mass/mass? give an example.

(viii) Define solubility.

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) Define oxidation and reduction.

(ii) What is meant by weak electrolytes? Give an example.

(iii) Find out the oxidation number of chlorin in $KClO_3$

(iv) Define corrosion.

(v) Write any two physical properties of metals.

(vi) What are transition metals?

(vii) Define electropositivity.

(viii) Write the reaction of chlorine with hot conc. NaOH.

> Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

> حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 09 نمبر ہیں۔

Q5(a). (Marks 05) Write any five differences between Rutherford's Atomic Theory and Bohr's Atomic Theory.

Q5(b). (Marks 04) Define and classify mixture with examples.

Q6(a). (Marks 05) Discuss ionic bond with example of NaCl.

Q6(b). (Marks 04) What is Boyle's Law? Derive its equation.

Q7(a). (Marks 05) Write any five rules for assigning oxidation number.

Q7(b). (Marks 04) Define mole. How 1M solution of NaOH is prepared?